In [ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Loading Dataset as Dataframe

In [ ]:
```
df = pd.read_csv('/content/DA dataset (1).csv')
```

In [ ]:
```
df.head()
df.shape
df.columns
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 600 entries, 0 to 599
Data columns (total 19 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   Patient_ID               600 non-null    object 
 1   Age                      600 non-null    int64  
 2   Gender                   600 non-null    object 
 3   Region                   600 non-null    object 
 4   Socioeconomic_Status     600 non-null    object 
 5   Occupation               600 non-null    object 
 6   Visit_Date               600 non-null    object 
 7   Symptoms                 600 non-null    object 
 8   Primary_Diagnosis        600 non-null    object 
 9   Blood_Glucose_mg_dL      600 non-null    float64
 10  HbA1c_%                  585 non-null    float64
 11  Total_Cholesterol_mg_dL  600 non-null    float64
 12  Treatment_Type           600 non-null    object 
 13  Treatment_Outcome        585 non-null    object 
 14  Imaging_Type             400 non-null    object 
 15  Imaging_Findings         600 non-null    object 
 16  Hospital_Type            600 non-null    object 
 17  Insurance_Covered        600 non-null    bool   
 18  BMI                      585 non-null    float64
dtypes: bool(1), float64(4), int64(1), object(13)
memory usage: 85.1+ KB
```

Dealing with missing values

In [ ]:
```
df.isnull().sum()
```

Out[243]:

	0
Patient_ID	0
Age	0
Gender	0
Region	0
Socioeconomic_Status	0
Occupation	0
Visit_Date	0
Symptoms	0
Primary_Diagnosis	0
Blood_Glucose_mg_dL	0
HbA1c_%	15
Total_Cholesterol_mg_dL	0
Treatment_Type	0
Treatment_Outcome	15
Imaging_Type	200
Imaging_Findings	0
Hospital_Type	0
Insurance_Covered	0
BMI	15

dtype: int64

Dealing with missing values in Imaging_Type Attribute

In [ ]:
```
df['Imaging_Type'].unique()
```

Out[244]:
```
array([nan, 'Ultrasound', 'CT Scan', 'X-Ray', 'MRI'], dtype=object)
```

In [ ]:
```
df['Imaging_Findings'].unique()
```

Out[245]:
```
array(['No imaging performed', 'Fatty liver changes',
       'No significant abnormality detected',
       'Mild cardiomegaly observed', 'Normal study',
       'Evidence of pleural effusion'], dtype=object)
```

In [ ]:
```
df.loc[df["Imaging_Type"].isnull(), "Imaging_Findings"]
```

Out[246]:

	Imaging_Findings
0	No imaging performed
3	No imaging performed
4	No imaging performed
6	No imaging performed
8	No imaging performed
...	...
590	No imaging performed
595	No imaging performed
596	No imaging performed
597	No imaging performed
599	No imaging performed

200 rows × 1 columns

dtype: object

In [ ]:
```
df.loc[df["Imaging_Type"].isnull(), "Imaging_Findings"].value_counts()
```

Out[247]:

	count
Imaging_Findings	
No imaging performed	200

dtype: int64

In [ ]:
```
df['Imaging_Type']=df['Imaging_Type'].fillna('No imaging performed')
df['Imaging_Type'].isnull()
```

Out[248]:

	Imaging_Type
0	False
1	False
2	False
3	False
4	False
...	...
595	False
596	False
597	False
598	False
599	False

600 rows × 1 columns

dtype: bool

Dealing with missing values in HbA1c_% Attribute

In [ ]:
```
df['HbA1c_%'].isnull().sum()
```

Out[249]:
```
np.int64(15)
```

In [ ]:
```
df.loc[df['HbA1c_%'].isnull(),'Primary_Diagnosis']
```

Out[250]:

	Primary_Diagnosis
84	Cardiovascular Disease
140	Type 2 Diabetes
154	Type 2 Diabetes
180	Cardiovascular Disease
236	Malaria
237	Cardiovascular Disease
238	Hypertension
284	Type 2 Diabetes
285	Hypertension
367	Hypertension
389	Hypertension
425	Chronic Kidney Disease
440	Cardiovascular Disease
562	Tuberculosis
563	Tuberculosis

dtype: object

In [ ]:
```
df['HbA1c_%'].describe()
```

Out[251]:

	HbA1c_%
count	585.000000
mean	6.423761
std	1.822096
min	2.200000
25%	5.100000
50%	6.100000
75%	7.400000
max	13.800000

dtype: float64

In [ ]:
```
df['HbA1c_%']=df['HbA1c_%'].fillna(df['HbA1c_%'].median())
df['HbA1c_%'].isnull()
```

Out[252]:

	HbA1c_%
0	False
1	False
2	False
3	False
4	False
...	...
595	False
596	False
597	False
598	False
599	False

600 rows × 1 columns

dtype: bool

Dealing with missing values in BMI Attribute

In [ ]:
```
df['BMI'].describe()
```

Out[253]:

	BMI
count	585.000000
mean	25.053333
std	7.172101
min	15.000000
25%	20.800000
50%	24.300000
75%	28.500000
max	72.500000

dtype: float64

In [ ]:
```
df.loc[df['BMI'].isnull(),['Age','Gender','Primary_Diagnosis','Symptoms']]
```

Out[254]:

	Age	Gender	Primary_Diagnosis	Symptoms
60	56	MALE	Malaria	Swelling in legs, high BP
82	44	Male	Thyroid Disorder	Fever, fatigue, increased thirst
119	52	Female	Type 2 Diabetes	Fever, fatigue, increased thirst
132	50	Male	Chronic Kidney Disease	Frequent urination, blurred vision
170	44	Male	Type 2 Diabetes	Fever, fatigue, increased thirst
172	37	Female	Dengue	Swelling in legs, high BP
223	29	Female	Thyroid Disorder	Swelling in legs, high BP
253	64	Female	Type 2 Diabetes	Headache, joint pain, high fever
294	14	Female	hypertension	Cough, weight loss, night sweats
400	36	Male	Type 2 Diabetes	Swelling in legs, high BP
407	45	Female	Hypertension	Cough, weight loss, night sweats
411	73	Female	Type 2 Diabetes	Chest pain, shortness of breath
459	46	Male	Dengue	Swelling in legs, high BP
505	54	Male	Malaria	Chest pain, shortness of breath
508	40	Female	cardiovascular disease	Frequent urination, blurred vision

In [ ]:
```
df['BMI']=df['BMI'].fillna(df['BMI'].median())
df['BMI'].isnull()
```

Out[255]:

	BMI
0	False
1	False
2	False
3	False
4	False
...	...
595	False
596	False
597	False
598	False
599	False

600 rows × 1 columns

dtype: bool

Dealing with missing values in Treatment_Outcome Attribute

In [ ]:
```
df['Treatment_Outcome'].unique()
```

Out[256]:
```
array(['Recovered', 'Worsened', 'Referred', nan, 'Improved', 'Stable'],
      dtype=object)
```

In [ ]:
```
df.loc[df['Treatment_Outcome'].isnull(),
['Treatment_Type','Primary_Diagnosis','Hospital_Type']]
```

Out[257]:

	Treatment_Type	Primary_Diagnosis	Hospital_Type
4	Surgery	Chronic Kidney Disease	Corporate
12	Combination Therapy	Type 2 Diabetes	Government
55	Surgery	Cardiovascular Disease	Government
128	Antibiotics	Cardiovascular Disease	Government
149	Insulin Therapy	Asthma/COPD	Corporate
186	Surgery	Cardiovascular Disease	Private
269	Surgery	Type 2 Diabetes	Corporate
288	Lifestyle Changes	Hypertension	Government
326	Surgery	Type 2 Diabetes	Government
378	Surgery	Type 2 Diabetes	Private
429	Medication	Type 2 Diabetes	Government
496	Medication	Dengue	Government
527	Insulin Therapy	Chronic Kidney Disease	Private
547	Antibiotics	Dengue	Government
579	Antibiotics	Dengue	Government

In [ ]:
```
df['Treatment_Outcome'].value_counts()
```

Out[258]:

Treatment_Outcome	count
Referred	129
Worsened	119
Recovered	115
Improved	113
Stable	109

dtype: int64

In [ ]:
```
df['Treatment_Outcome']=df['Treatment_Outcome'].fillna('Unknown')
df['Treatment_Outcome'].isnull()
```

Out[259]:

	Treatment_Outcome
0	False
1	False
2	False
3	False
4	False
...	...
595	False
596	False
597	False
598	False
599	False

600 rows × 1 columns

dtype: bool

Handling Duplicate values

In [ ]:
```
df.duplicated().value_counts()
```

Out[260]:

	count
False	590
True	10

dtype: int64

In [ ]:
```
df[df.duplicated()]
```

Out[261]: Columns 1–8 of 20

	Patient_ID	Age	Gender	Region	Socioeconomic_Status	Occupation	Visit_Date
590	72A64225-C9A	63	Female	Andhra Pradesh	Middle	Clinical cytogenetic ist	2025-03-11
591	BBD2BA91-057	31	Male	Kerala	Middle	Brewing technologis t	2024-07-07
592	7941194D-9DF	26	Female	Delhi	Middle	Advertising art director	2024-09-27
593	537F741A-D20	33	Male	West Bengal	High	Farm manager	2025-04-08
594	392CBC4A-4D3	76	Male	Puducherry	Low	Operations geologist	2025-04-13
595	669C6BE4-4FF	52	Male	Uttar Pradesh	Low	Games developer	2024-12-26
596	4B1CAB21-769	43	Female	Tamil Nadu	Low	Merchandis er, retail	2025-01-26
597	1E05D9EC-062	26	Female	Andhra Pradesh	Middle	Analytical chemist	2024-06-15
598	8407A138-01A	41	Male	Tamil Nadu	High	Intelligence analyst	2024-11-05
599	9A82F1B8-772	67	Male	West Bengal	Middle	Journalist, newspaper	2026-03-18

Columns 9–15 of 20 · first column repeated

	Symptoms	Primary_Diagnosis	Blood_Glucose_mg_dL	HbA1c_%	Total_Cholesterol_mg_dL	Treatment_Type	Treatment_Outcome
590	Cough, weight loss, night sweats	Other	62.9	5.1	180.2	Surgery	Worsened
591	Frequent urination, blurred vision	Hypertensio n	149.9	4.4	331.6	Antibiotics	Referred
592	Headache, joint pain, high fever	Malaria	109.9	6.2	205.8	Physiothera py	Stable
593	Cough, weight loss, night sweats	Type 2 Diabetes	148.3	10.4	225.6	Surgery	Stable
594	Frequent urination, blurred vision	Thyroid Disorder	136.7	8.0	209.5	Combinatio n Therapy	Improved

	Symptoms	Primary_Diagnosis	Blood_Glucose_mg_dL	HbA1c_%	Total_Cholesterol_mg_dL	Treatment_Type	Treatment_Outcome
595	Cough, weight loss, night sweats	Chronic Kidney Disease	60.7	3.4	152.8	Antibiotics	Worsened
596	Fever, fatigue, increased thirst	type 2 diabetes	157.6	8.7	186.2	Antibiotics	Referred
597	Frequent urination, blurred vision	Type 2 Diabetes	175.1	10.1	249.4	Antibiotics	Improved
598	Cough, weight loss, night sweats	chronic kidney disease	78.8	4.8	222.4	Combination Therapy	Referred
599	Headache, joint pain, high fever	Dengue	129.6	8.2	188.3	Physiotherapy	Worsened

Columns 16–20 of 20 · first column repeated

	Imaging_Type	Imaging_Findings	Hospital_Type	Insurance_Covered	BMI
590	No imaging performed	No imaging performed	Corporate	True	22.7
591	Ultrasound	Mild cardiomegaly observed	Private	False	28.2
592	Ultrasound	Normal study	Government	False	24.6
593	X-Ray	Evidence of pleural effusion	Corporate	False	29.7
594	X-Ray	Evidence of pleural effusion	Government	False	22.1
595	No imaging performed	No imaging performed	Government	False	34.0
596	No imaging performed	No imaging performed	Private	True	20.8
597	No imaging performed	No imaging performed	Government	False	72.5
598	Ultrasound	No significant abnormality detected	Private	True	32.3
599	No imaging performed	No imaging performed	Private	False	15.8

In [ ]:
```
df['Patient_ID'].nunique()
```

Out[262]:
```
590
```

In [ ]:
```
df["Patient_ID"].value_counts()[df["Patient_ID"].value_counts() > 1]
```

Out[263]:

Patient_ID	count
669C6BE4-4FF	2
392CBC4A-4D3	2
1E05D9EC-062	2
BBD2BA91-057	2
9A82F1B8-772	2
72A64225-C9A	2
4B1CAB21-769	2
8407A138-01A	2
7941194D-9DF	2
537F741A-D20	2

dtype: int64

In [ ]:
```
df[df["Patient_ID"].duplicated(keep=False)].sort_values("Patient_ID")
```

Out[264]: Columns 1–8 of 20

	Patient_ID	Age	Gender	Region	Socioeconomic_Status	Occupation	Visit_Date
597	1E05D9EC-062	26	Female	Andhra Pradesh	Middle	Analytical chemist	2024-06-15
372	1E05D9EC-062	26	Female	Andhra Pradesh	Middle	Analytical chemist	2024-06-15
152	392CBC4A-4D3	76	Male	Puducherry	Low	Operations geologist	2025-04-13
594	392CBC4A-4D3	76	Male	Puducherry	Low	Operations geologist	2025-04-13
596	4B1CAB21-769	43	Female	Tamil Nadu	Low	Merchandiser, retail	2025-01-26
355	4B1CAB21-769	43	Female	Tamil Nadu	Low	Merchandiser, retail	2025-01-26
385	537F741A-D20	33	Male	West Bengal	High	Farm manager	2025-04-08
593	537F741A-D20	33	Male	West Bengal	High	Farm manager	2025-04-08
258	669C6BE4-4FF	52	Male	Uttar Pradesh	Low	Games developer	2024-12-26
595	669C6BE4-4FF	52	Male	Uttar Pradesh	Low	Games developer	2024-12-26
26	72A64225-C9A	63	Female	Andhra Pradesh	Middle	Clinical cytogeneticist	2025-03-11
590	72A64225-C9A	63	Female	Andhra Pradesh	Middle	Clinical cytogeneticist	2025-03-11
212	7941194D-9DF	26	Female	Delhi	Middle	Advertising art director	2024-09-27
592	7941194D-9DF	26	Female	Delhi	Middle	Advertising art director	2024-09-27
485	8407A138-01A	41	Male	Tamil Nadu	High	Intelligence analyst	2024-11-05
598	8407A138-01A	41	Male	Tamil Nadu	High	Intelligence analyst	2024-11-05
125	9A82F1B8-772	67	Male	West Bengal	Middle	Journalist, newspaper	2026-03-18
599	9A82F1B8-772	67	Male	West Bengal	Middle	Journalist, newspaper	2026-03-18
72	BBD2BA91-057	31	Male	Kerala	Middle	Brewing technologist	2024-07-07
591	BBD2BA91-057	31	Male	Kerala	Middle	Brewing technologist	2024-07-07

Columns 9–15 of 20 · first column repeated

	Symptoms	Primary_Diagnosis	Blood_Glucose_mg_dL	HbA1c_%	Total_Cholesterol_mg_dL	Treatment_Type	Treatment_Outcome
597	Frequent urination, blurred vision	Type 2 Diabetes	175.1	10.1	249.4	Antibiotics	Improved
372	Frequent urination, blurred vision	Type 2 Diabetes	175.1	10.1	249.4	Antibiotics	Improved
152	Frequent urination, blurred vision	Thyroid Disorder	136.7	8.0	209.5	Combination Therapy	Improved
594	Frequent urination, blurred vision	Thyroid Disorder	136.7	8.0	209.5	Combination Therapy	Improved
596	Fever, fatigue, increased thirst	type 2 diabetes	157.6	8.7	186.2	Antibiotics	Referred
355	Fever, fatigue, increased thirst	type 2 diabetes	157.6	8.7	186.2	Antibiotics	Referred
385	Cough, weight loss, night sweats	Type 2 Diabetes	148.3	10.4	225.6	Surgery	Stable
593	Cough, weight loss, night sweats	Type 2 Diabetes	148.3	10.4	225.6	Surgery	Stable
258	Cough, weight loss, night sweats	Chronic Kidney Disease	60.7	3.4	152.8	Antibiotics	Worsened
595	Cough, weight loss, night sweats	Chronic Kidney Disease	60.7	3.4	152.8	Antibiotics	Worsened
26	Cough, weight loss, night sweats	Other	62.9	5.1	180.2	Surgery	Worsened
590	Cough, weight loss, night sweats	Other	62.9	5.1	180.2	Surgery	Worsened

	Symptoms	Primary_Diagnosis	Blood_Glucose_mg_dL	HbA1c_%	Total_Cholesterol_mg_dL	Treatment_Type	Treatment_Outcome
212	Headache, joint pain, high fever	Malaria	109.9	6.2	205.8	Physiotherapy	Stable
592	Headache, joint pain, high fever	Malaria	109.9	6.2	205.8	Physiotherapy	Stable
485	Cough, weight loss, night sweats	chronic kidney disease	78.8	4.8	222.4	Combination Therapy	Referred
598	Cough, weight loss, night sweats	chronic kidney disease	78.8	4.8	222.4	Combination Therapy	Referred
125	Headache, joint pain, high fever	Dengue	129.6	8.2	188.3	Physiotherapy	Worsened
599	Headache, joint pain, high fever	Dengue	129.6	8.2	188.3	Physiotherapy	Worsened
72	Frequent urination, blurred vision	Hypertension	149.9	4.4	331.6	Antibiotics	Referred
591	Frequent urination, blurred vision	Hypertension	149.9	4.4	331.6	Antibiotics	Referred

Columns 16–20 of 20 · first column repeated

	Imaging_Type	Imaging_Findings	Hospital_Type	Insurance_Covered	BMI
597	No imaging performed	No imaging performed	Government	False	72.5
372	No imaging performed	No imaging performed	Government	False	72.5
152	X-Ray	Evidence of pleural effusion	Government	False	22.1
594	X-Ray	Evidence of pleural effusion	Government	False	22.1
596	No imaging performed	No imaging performed	Private	True	20.8
355	No imaging performed	No imaging performed	Private	True	20.8
385	X-Ray	Evidence of pleural effusion	Corporate	False	29.7
593	X-Ray	Evidence of pleural effusion	Corporate	False	29.7

	Imaging_Type	Imaging_Findings	Hospital_Type	Insurance_Covered	BMI
258	No imaging performed	No imaging performed	Government	False	34.0
595	No imaging performed	No imaging performed	Government	False	34.0
26	No imaging performed	No imaging performed	Corporate	True	22.7
590	No imaging performed	No imaging performed	Corporate	True	22.7
212	Ultrasound	Normal study	Government	False	24.6
592	Ultrasound	Normal study	Government	False	24.6
485	Ultrasound	No significant abnormality detected	Private	True	32.3
598	Ultrasound	No significant abnormality detected	Private	True	32.3
125	No imaging performed	No imaging performed	Private	False	15.8
599	No imaging performed	No imaging performed	Private	False	15.8
72	Ultrasound	Mild cardiomegaly observed	Private	False	28.2
591	Ultrasound	Mild cardiomegaly observed	Private	False	28.2

In [ ]:
```
df = df.drop_duplicates()
df.shape
df['Patient_ID'].nunique()
```

Out[265]:
```
590
```

Data Consistency

In [ ]:
```
df['Primary_Diagnosis'].value_counts()
```

Out[266]:

Primary_Diagnosis	count
Type 2 Diabetes	166
Hypertension	126
Cardiovascular Disease	65
Dengue	49
Chronic Kidney Disease	41
Tuberculosis	29
Malaria	29
Asthma/COPD	24
Thyroid Disorder	23
type 2 diabetes	11
hypertension	8
Other	5
tuberculosis	3
chronic kidney disease	3
cardiovascular disease	3
dengue	2
asthma/copd	1
malaria	1
thyroid disorder	1

dtype: int64

In [ ]:
```
mapping = {
    "hypertension":"Hypertension",
    "Hypertension":"Hypertension",
    "type 2 diabetes":"Type 2 Diabetes",
    "Type 2 Diabetes":"Type 2 Diabetes",
    "chronic kidney disease":"Chronic Kidney Disease",
    "Chronic Kidney Disease":"Chronic Kidney Disease",
    "cardiovascular disease":"Cardiovascular Disease",
    "Cardiovascular Disease":"Cardiovascular Disease",
    "tuberculosis":"Tuberculosis",
    "Tuberculosis":"Tuberculosis",
    "dengue":"Dengue",
    "Dengue":"Dengue",
    "thyroid disorder":"Thyroid Disorder",
    "Thyroid Disorder":"Thyroid Disorder",
    "malaria":"Malaria",
    "Malaria":"Malaria",
    "asthma/copd":"Asthma/COPD",
    "Asthma/COPD":"Asthma/COPD"}
df['Primary_Diagnosis'] = df['Primary_Diagnosis'].map(mapping)
```

In [ ]:
```
df['Primary_Diagnosis'].value_counts()
```

Out[268]:

Primary_Diagnosis	count
Type 2 Diabetes	166
Hypertension	126
Cardiovascular Disease	65
Dengue	49
Chronic Kidney Disease	41
Malaria	29
Tuberculosis	29
Asthma/COPD	24
Thyroid Disorder	23

dtype: int64

In [ ]:
```
df['Gender'].value_counts()
```

Out[269]:

Gender	count
Female	285
Male	266
Female	9
Male	7
MALE	7
female	6
male	4
FEMALE	4
Other	2

dtype: int64

In [ ]:
```
maped = {
    'FEMALE':'Female',
    'Female':'Female',
    'female':'Female',
    'male':'Male',
    'Male':'Male',
    'MALE':'Male',
    'Other':'Other'
}
df['Gender'] = df['Gender'].map(maped)
```

In [ ]:
```
df['Gender'].value_counts()
```

Out[271]:

Gender	count
Female	295
Male	277
Other	2

dtype: int64

In [ ]:
```
df['Region'].value_counts()
```

Out[272]:

Region	count
Tamil Nadu	117
Kerala	82
Karnataka	75
Andhra Pradesh	58
Delhi	55
Telangana	49
Puducherry	44
Uttar Pradesh	43
Maharashtra	35
West Bengal	32

dtype: int64

In [ ]:
```
df['Insurance_Covered'].value_counts()
```

Out[273]:

Insurance_Covered	count
False	375
True	215

dtype: int64

In [ ]:
```
df['Treatment_Type'].value_counts()
```

Out[274]:

Treatment_Type	count
Antibiotics	94
Lifestyle Changes	91
Surgery	85
Insulin Therapy	82
Medication	81
Combination Therapy	79
Physiotherapy	78

dtype: int64

In [ ]:
```
df['Treatment_Outcome'].value_counts()
```

Out[275]:

Treatment_Outcome	count
Referred	126
Worsened	116
Recovered	115
Improved	111
Stable	107
Unknown	15

dtype: int64

In [ ]:
```
df['Imaging_Type'].value_counts()
```

Out[276]:

Imaging_Type	count
No imaging performed	195
X-Ray	163
Ultrasound	119
CT Scan	71
MRI	42

dtype: int64

In [ ]:
```
df['Hospital_Type'].value_counts()
```

Out[277]:

Hospital_Type	count
Government	285
Private	222
Corporate	83

dtype: int64

In [ ]:
```
df['Socioeconomic_Status'].value_counts()
```

Out[278]:

Socioeconomic_Status	count
Middle	268
Low	238
High	84

dtype: int64

In [ ]:
```
df['Occupation'].head(11)
```

Out[279]:

	Occupation
0	IT sales professional
1	Community pharmacist
2	Tourism officer
3	Mental health nurse
4	Operations geologist
5	Engineer, agricultural
6	Pharmacist, community
7	Training and development officer
8	Office manager
9	Hydrogeologist
10	Air broker

dtype: object

In [ ]:
```
df['Age'].describe()
```

Out[280]:

	Age
count	590.000000
mean	44.166102
std	17.877138
min	1.000000
25%	32.000000
50%	44.000000
75%	57.000000
max	94.000000

dtype: float64

In [ ]:
```
df['Blood_Glucose_mg_dL'].describe()
```

Out[281]:

	Blood_Glucose_mg_dL
count	590.000000
mean	138.118644
std	92.939512
min	34.500000
25%	96.425000
50%	122.500000
75%	161.425000
max	999.000000

dtype: float64

In [ ]:
```
df[df['Blood_Glucose_mg_dL'] ≠ 999].describe()
```

Out[282]:

	Age	Blood_Glucose_mg_dL	HbA1c_%	Total_Cholesterol_mg_dL	BMI
count	585.000000	585.000000	585.000000	585.000000	585.000000
mean	44.261538	130.760684	6.400855	217.675897	24.933504
std	17.872311	48.086346	1.790055	43.425947	6.832036
min	1.000000	34.500000	2.200000	67.300000	15.000000
25%	32.000000	96.300000	5.200000	189.400000	20.900000
50%	44.000000	122.100000	6.100000	215.700000	24.300000
75%	58.000000	159.300000	7.400000	245.100000	28.300000
max	94.000000	335.900000	13.800000	335.500000	72.500000

In [ ]:
```
df["Blood_Glucose_mg_dL"] = df["Blood_Glucose_mg_dL"].replace(999, np.nan)
```

In [ ]:
```
df["Blood_Glucose_mg_dL"].isnull().sum()
```

Out[284]:
```
np.int64(5)
```

In [ ]:
```
df['Blood_Glucose_mg_dL'].median()
```

Out[285]:
```
122.1
```

In [ ]:
```
df['Bloo_Glucose_mg_dL']=df['Blood_Glucose_mg_dL'].fillna(df['Blood_Glucose_mg_dL'].median())
```

In [ ]:
```
df['Bloo_Glucose_mg_dL'].isnull().sum()
```

Out[287]:
```
np.int64(0)
```

In [ ]:
```
df['HbA1c_%'].describe()
```

Out[288]:

	HbA1c_%
count	590.000000
mean	6.406949
std	1.787754
min	2.200000
25%	5.200000
50%	6.100000
75%	7.400000
max	13.800000

dtype: float64

In [ ]:
```
df['Total_Cholesterol_mg_dL'].describe()
```

Out[289]:

	Total_Cholesterol_mg_dL
count	590.000000
mean	217.692712
std	43.348534
min	67.300000
25%	189.425000
50%	215.600000
75%	245.025000
max	335.500000

dtype: float64

In [ ]:
```
df['BMI'].describe()
```

Out[290]:

	BMI
count	590.000000
mean	24.945763
std	6.834034
min	15.000000
25%	20.900000
50%	24.300000
75%	28.300000
max	72.500000

dtype: float64

In [ ]:
```
df['Gender'].isnull().sum()
```

Out[291]:
```
np.int64(16)
```

In [ ]:
```
original_df = pd.read_csv("/content/DA dataset (1).csv")
original_df = original_df.drop_duplicates()
gender_map = original_df.set_index("Patient_ID")["Gender"]
df["Gender"] = df["Patient_ID"].map(gender_map)
df["Gender"] = df["Gender"].replace({
    "FEMALE": "Female",
    "female": "Female",
    "MALE": "Male",
    "male": "Male"
})
if "Bloo_Glucose_mg_dL" in df.columns:
    df.drop(columns=["Bloo_Glucose_mg_dL"], inplace=True)
print("Missing Gender:", df["Gender"].isnull().sum())
print("\nGender counts:")
print(df["Gender"].value_counts())

print("\nExtra glucose column present:",
      "Bloo_Glucose_mg_dL" in df.columns)
```

```
Missing Gender: 0

Gender counts:
Gender
Female      295
Male        277
 Female       9
 Male         7
Other         2
Name: count, dtype: int64

Extra glucose column present: False
```

In [ ]:
```
df["Gender"] = df["Gender"].str.strip()
df["Gender"] = df["Gender"].replace({
    "FEMALE": "Female",
    "female": "Female",
    "MALE": "Male",
    "male": "Male"
})
print("Missing Gender:", df["Gender"].isnull().sum())
print("\nGender counts:")
print(df["Gender"].value_counts())

print("\nTotal rows:", len(df))
```

```
Missing Gender: 0

Gender counts:
Gender
Female    304
Male      284
Other       2
Name: count, dtype: int64

Total rows: 590
```

Outliers detection using IQR method

In [ ]:
```
# finding outliers in BMI
q1 = df['BMI'].quantile(0.25)
q3 = df['BMI'].quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5*iqr
upper = q3 + 1.5*iqr

outliers = df[(df['BMI'] < lower) | (df['BMI'] > upper)]
print(outliers)
```

```
       Patient_ID  Age  Gender          Region Socioeconomic_Status  \
103  4393837F-64D   46  Female           Delhi                  Low
154  6D87AB92-2C2   81    Male      Tamil Nadu               Middle
372  1E05D9EC-062   26  Female  Andhra Pradesh               Middle
549  93DA27B6-B52   22    Male       Karnataka               Middle
580  15F99EB1-F15   60    Male           Delhi                  Low

                          Occupation  Visit_Date  \
103            Chief Strategy Officer  2026-01-18
154                Materials engineer  2025-03-26
372                Analytical chemist  2024-06-15
549  Sales promotion account executive  2024-11-08
580           Insurance risk surveyor  2025-08-01

                            Symptoms Primary_Diagnosis  \
103    Chest pain, shortness of breath   Type 2 Diabetes
154   Fever, fatigue, increased thirst   Type 2 Diabetes
372  Frequent urination, blurred vision   Type 2 Diabetes
549   Fever, fatigue, increased thirst   Type 2 Diabetes
580    Chest pain, shortness of breath   Type 2 Diabetes

     Blood_Glucose_mg_dL  HbA1c_%  Total_Cholesterol_mg_dL Treatment_Type  \
103                165.2      7.6                   283.6  Physiotherapy
154                212.2      6.1                   239.9     Medication
372                175.1     10.1                   249.4    Antibiotics
549                209.9      6.1                   229.5    Antibiotics
580                335.9      8.6                   247.8  Physiotherapy

    Treatment_Outcome          Imaging_Type  \
103         Recovered               CT Scan
154         Recovered                 X-Ray
372          Improved  No imaging performed
549          Worsened             Ultrasound
580         Recovered                   MRI

                       Imaging_Findings Hospital_Type  Insurance_Covered  \
103         Evidence of pleural effusion    Government               True
154  No significant abnormality detected     Corporate               True
372                 No imaging performed    Government              False
549         Evidence of pleural effusion    Government              False
580           Mild cardiomegaly observed    Government              False

      BMI
```

```
103  72.5
154  72.5
372  72.5
549  72.5
580  72.5
```

In [ ]:
```
# finding outliers in Blood_Glucose_mg_dL
q1 = df['Blood_Glucose_mg_dL'].quantile(0.25)
q3 = df['Blood_Glucose_mg_dL'].quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5*iqr
upper = q3 + 1.5*iqr

outliers = df[(df['Blood_Glucose_mg_dL'] < lower) |
(df['Blood_Glucose_mg_dL'] > upper)]
print(outliers)
```

```
       Patient_ID  Age  Gender          Region Socioeconomic_Status  \
62   E87432C8-0A9   42  Female           Delhi                  Low
183  70C07989-0A7   58  Female   Uttar Pradesh               Middle
187  4B85DEB7-1DC   59    Male       Karnataka                 High
227  A7E8F702-DD7   78    Male   Uttar Pradesh                  Low
411  4543F031-183   73  Female   Uttar Pradesh                  Low
427  57050C8A-E5E   59  Female   Uttar Pradesh                 High
457  24089CE8-F9C   49    Male  Andhra Pradesh               Middle
477  5D36CC6A-44D   49    Male     West Bengal                  Low
565  63B4D48C-6CB   45  Female      Tamil Nadu                  Low
580  15F99EB1-F15   60    Male           Delhi                  Low

                                 Occupation  Visit_Date  \
62                           Mining engineer  2024-07-27
183  Geographical information systems officer  2025-05-31
187                            Administrator  2026-03-13
227        Tourist information centre manager  2025-12-30
411                      Newspaper journalist  2026-03-19
427                           Office manager  2024-11-11
457                    Glass blower/designer  2025-04-05
477                            Hotel manager  2025-12-29
565                                      Sub  2024-09-19
580                  Insurance risk surveyor  2025-08-01

                              Symptoms Primary_Diagnosis  \
62   Frequent urination, blurred vision               NaN
183  Frequent urination, blurred vision   Type 2 Diabetes
187    Fever, fatigue, increased thirst   Type 2 Diabetes
227    Headache, joint pain, high fever   Type 2 Diabetes
411     Chest pain, shortness of breath   Type 2 Diabetes
427    Fever, fatigue, increased thirst   Type 2 Diabetes
457    Fever, fatigue, increased thirst   Type 2 Diabetes
477    Headache, joint pain, high fever   Type 2 Diabetes
565    Fever, fatigue, increased thirst   Type 2 Diabetes
580     Chest pain, shortness of breath   Type 2 Diabetes

     Blood_Glucose_mg_dL  HbA1c_%  Total_Cholesterol_mg_dL  \
62                 258.3     11.7                   211.1
183                257.8      8.7                   218.7
187                281.5      9.9                   229.0
227                256.0      8.8                   188.1
411                276.4      7.6                   140.5
```

```
427               269.4      8.7                   171.3
457               314.3      8.2                   194.2
477               275.3      8.2                   175.7
565               269.0      3.1                   213.4
580               335.9      8.6                   247.8

         Treatment_Type Treatment_Outcome Imaging_Type  \
62           Medication          Referred          MRI
183   Lifestyle Changes          Worsened   Ultrasound
187          Medication            Stable        X-Ray
227             Surgery          Improved          MRI
411   Lifestyle Changes         Recovered   Ultrasound
427     Insulin Therapy          Referred        X-Ray
457   Lifestyle Changes         Recovered          MRI
477  Combination Therapy         Recovered      CT Scan
565          Medication          Worsened   Ultrasound
580       Physiotherapy         Recovered          MRI

                      Imaging_Findings Hospital_Type  Insurance_Covered  \
62   No significant abnormality detected    Government              False
183                 Fatty liver changes       Private              False
187        Evidence of pleural effusion       Private              False
227        Evidence of pleural effusion       Private              False
411  No significant abnormality detected    Government              False
427                        Normal study    Government              False
457          Mild cardiomegaly observed       Private              False
477          Mild cardiomegaly observed     Corporate               True
565                Fatty liver changes       Private              False
580          Mild cardiomegaly observed    Government              False

      BMI
62   27.8
183  19.7
187  22.9
227  28.5
411  24.3
427  29.3
457  17.8
477  28.6
565  22.1
580  72.5
```

In [ ]:
```
# finding outliers in HbA1c
q1 = df['HbA1c_%'].quantile(0.25)
q3 = df['HbA1c_%'].quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5*iqr
upper = q3 + 1.5*iqr

outliers = df[(df['HbA1c_%'] < lower) | (df['HbA1c_%'] > upper)]
print(outliers)
```

```
      Patient_ID  Age  Gender          Region Socioeconomic_Status  \
11   E6E909A7-6B8   39    Male          Kerala                  Low
62   E87432C8-0A9   42  Female           Delhi                  Low
89   4B81A0FC-A4C   58    Male  Andhra Pradesh                 High
92   EF9BF960-052   66    Male     Maharashtra                  Low
98   DC84808D-AF8   52  Female          Kerala                 High
119  B01DA2CA-F91   52  Female      Puducherry                  Low
170  A449E219-24D   44    Male     West Bengal                  Low
231  9BAB9A3D-A6F   64    Male           Delhi               Middle
302  CCD004B3-795   57  Female      Puducherry                  Low
331  7A345FA7-590   57  Female      Tamil Nadu               Middle
410  2F816B02-0B0   39    Male       Telangana               Middle
419  BD3E18C8-D69   34    Male   Uttar Pradesh                 High
455  F96962F5-BF6   37    Male      Tamil Nadu                  Low
466  29196AD7-120   45  Female          Kerala                  Low
513  1FB82F72-645   43  Female      Puducherry               Middle

                           Occupation  Visit_Date  \
11              Holiday representative  2026-01-10
62                    Mining engineer  2024-07-27
89                Art gallery manager  2025-08-30
92                       Immunologist  2024-07-24
98                   Graphic designer  2024-12-31
119          Learning disability nurse  2024-08-03
170              Community arts worker  2024-06-18
231                      Town planner  2024-08-31
302                Structural engineer  2025-11-27
331        Conservator, museum/gallery  2024-10-02
410                  Surveyor, mining  2026-04-15
419                     Herpetologist  2025-12-26
455  Public relations account executive  2026-04-22
466               Licensed conveyancer  2024-06-02
513                         Adult nurse  2025-08-28

                             Symptoms Primary_Diagnosis  \
11     Headache, joint pain, high fever   Type 2 Diabetes
62   Frequent urination, blurred vision              NaN
89            Swelling in legs, high BP   Type 2 Diabetes
92     Headache, joint pain, high fever   Type 2 Diabetes
98            Swelling in legs, high BP   Type 2 Diabetes
119    Fever, fatigue, increased thirst   Type 2 Diabetes
170    Fever, fatigue, increased thirst   Type 2 Diabetes
231     Chest pain, shortness of breath   Type 2 Diabetes
```

```
302  Frequent urination, blurred vision   Type 2 Diabetes
331    Headache, joint pain, high fever   Type 2 Diabetes
410     Chest pain, shortness of breath   Type 2 Diabetes
419    Fever, fatigue, increased thirst   Type 2 Diabetes
455    Fever, fatigue, increased thirst   Type 2 Diabetes
466    Fever, fatigue, increased thirst   Type 2 Diabetes
513    Headache, joint pain, high fever   Type 2 Diabetes

     Blood_Glucose_mg_dL  HbA1c_%  Total_Cholesterol_mg_dL  \
11                 120.6     10.8                   225.0
62                 258.3     11.7                   211.1
89                  86.0     12.2                   177.0
92                 150.1     13.6                   193.5
98                 175.2     12.0                   240.5
119                167.5     11.4                   219.2
170                165.4     11.1                   207.7
231                161.8     10.8                   260.6
302                117.0     10.8                   198.4
331                114.6     10.9                   222.5
410                168.3     11.2                   243.4
419                159.5     11.2                   178.0
455                185.4     10.9                   238.0
466                237.9     11.4                   145.0
513                207.5     13.8                   262.9

          Treatment_Type Treatment_Outcome          Imaging_Type  \
11       Insulin Therapy          Referred  No imaging performed
62            Medication          Referred                   MRI
89            Medication          Referred                 X-Ray
92     Lifestyle Changes          Referred  No imaging performed
98            Medication          Worsened             Ultrasound
119              Surgery          Worsened                 X-Ray
170          Antibiotics          Improved               CT Scan
231        Physiotherapy          Worsened                 X-Ray
302      Insulin Therapy            Stable                 X-Ray
331          Antibiotics          Referred  No imaging performed
410    Lifestyle Changes          Worsened                   MRI
419  Combination Therapy          Referred               CT Scan
455  Combination Therapy          Referred               CT Scan
466    Lifestyle Changes            Stable  No imaging performed
513      Insulin Therapy          Improved               CT Scan

                       Imaging_Findings Hospital_Type  Insurance_Covered  \
11                  No imaging performed    Government              False
62   No significant abnormality detected    Government              False
89             Mild cardiomegaly observed       Private              False
92                  No imaging performed       Private              False
98   No significant abnormality detected     Corporate              False
119  No significant abnormality detected    Government              False
170         Evidence of pleural effusion    Government              False
231            Mild cardiomegaly observed    Government              False
302         Evidence of pleural effusion       Private              False
331                 No imaging performed       Private              False
410  No significant abnormality detected    Government              False
```

419	Mild cardiomegaly observed	Corporate	False
455	Mild cardiomegaly observed	Government	False
466	No imaging performed	Private	True
513	Evidence of pleural effusion	Private	False

	BMI
11	31.8
62	27.8
89	15.0
92	19.7
98	17.4
119	24.3
170	24.3
231	23.4
302	30.8
331	30.4
410	27.1
419	26.5
455	23.6
466	15.8
513	18.3

In [ ]:
```
# finding outliers iTotal_Cholesterol_mg_dL
q1 = df['Total_Cholesterol_mg_dL'].quantile(0.25)
q3 = df['Total_Cholesterol_mg_dL'].quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5*iqr
upper = q3 + 1.5*iqr

outliers = df[(df['Total_Cholesterol_mg_dL'] < lower) |
(df['Total_Cholesterol_mg_dL'] > upper)]
print(outliers)
```

```
       Patient_ID  Age  Gender      Region Socioeconomic_Status  \
72   BBD2BA91-057   31    Male      Kerala                Middle
235  B4157E65-77D   29    Male   Karnataka                Middle
468  41FF5B07-F95   59    Male   Karnataka                   Low
557  4BB5F449-6F3   25  Female  Tamil Nadu                   Low
579  77DAAA23-FCD   10  Female  Tamil Nadu                Middle

                Occupation  Visit_Date                          Symptoms
\
72     Brewing technologist  2024-07-07  Frequent urination, blurred vision
235        Fashion designer  2025-07-18    Cough, weight loss, night sweats
468           Advice worker  2024-08-13           Swelling in legs, high BP
557  Pension scheme manager  2026-03-01           Swelling in legs, high BP
579            Set designer  2024-12-19           Swelling in legs, high BP

    Primary_Diagnosis  Blood_Glucose_mg_dL  HbA1c_%  Total_Cholesterol_mg_dL
\
72       Hypertension                149.9      4.4                    331.6
235       Asthma/COPD                133.3      7.8                     67.3
468      Tuberculosis                 61.8      4.9                    104.4
557      Hypertension                 76.9      5.3                    335.5
579            Dengue                142.1      5.8                     82.9

        Treatment_Type Treatment_Outcome          Imaging_Type  \
72         Antibiotics          Referred            Ultrasound
235  Lifestyle Changes            Stable  No imaging performed
468            Surgery         Recovered                 X-Ray
557         Antibiotics         Recovered                 X-Ray
579         Antibiotics           Unknown  No imaging performed

               Imaging_Findings Hospital_Type  Insurance_Covered   BMI
72   Mild cardiomegaly observed       Private              False  28.2
235        No imaging performed       Private              False  15.0
468                Normal study    Government               True  28.5
557                Normal study     Corporate              False  31.3
579        No imaging performed    Government              False  23.7
```

In [ ]:
```
df["Primary_Diagnosis"] = df["Primary_Diagnosis"].replace(
    r"^\s*$", np.nan, regex=True
)
df["Primary_Diagnosis"] = df["Primary_Diagnosis"].fillna(
    "Unknown / Not recorded"
)
print("Missing values after handling:")
print(df["Primary_Diagnosis"].isnull().sum())

print("\nDiagnosis-wise record counts:")
print(df["Primary_Diagnosis"].value_counts())
```

```
Missing values after handling:
0

Diagnosis-wise record counts:
Primary_Diagnosis
Type 2 Diabetes           166
Hypertension              126
Cardiovascular Disease     65
Dengue                     49
Chronic Kidney Disease     41
Unknown / Not recorded     38
Malaria                    29
Tuberculosis               29
Asthma/COPD                24
Thyroid Disorder           23
Name: count, dtype: int64
```

In [ ]:
```
df["Blood_Glucose_mg_dL"] = df["Blood_Glucose_mg_dL"].replace(
    999, np.nan
)
glucose_median = df["Blood_Glucose_mg_dL"].median()
df["Blood_Glucose_mg_dL"] = df["Blood_Glucose_mg_dL"].fillna(glucose_median)
print("Missing glucose values:", df["Blood_Glucose_mg_dL"].isnull().sum())
print("Median used:", glucose_median)
```

```
Missing glucose values: 0
Median used: 122.1
```

In [ ]:
```
missing_values = df.isnull().sum()
print("Missing values in each column:")
print(missing_values[missing_values > 0])
print("\nTotal missing values in dataset:",
      df.isnull().sum().sum())
```

```
Missing values in each column:
Series([], dtype: int64)

Total missing values in dataset: 0
```

In [ ]:
```
print("Total rows:", len(df))
print("Total columns:", len(df.columns))
print("Duplicate rows:", df.duplicated().sum())
print("Missing diagnosis:",
      df["Primary_Diagnosis"].isnull().sum())
print("Missing glucose:",
      df["Blood_Glucose_mg_dL"].isnull().sum())
print("Missing dates:",
      df["Visit_Date"].isnull().sum())
```

```
Total rows: 590
Total columns: 19
Duplicate rows: 0
Missing diagnosis: 0
Missing glucose: 0
Missing dates: 0
```

Insights

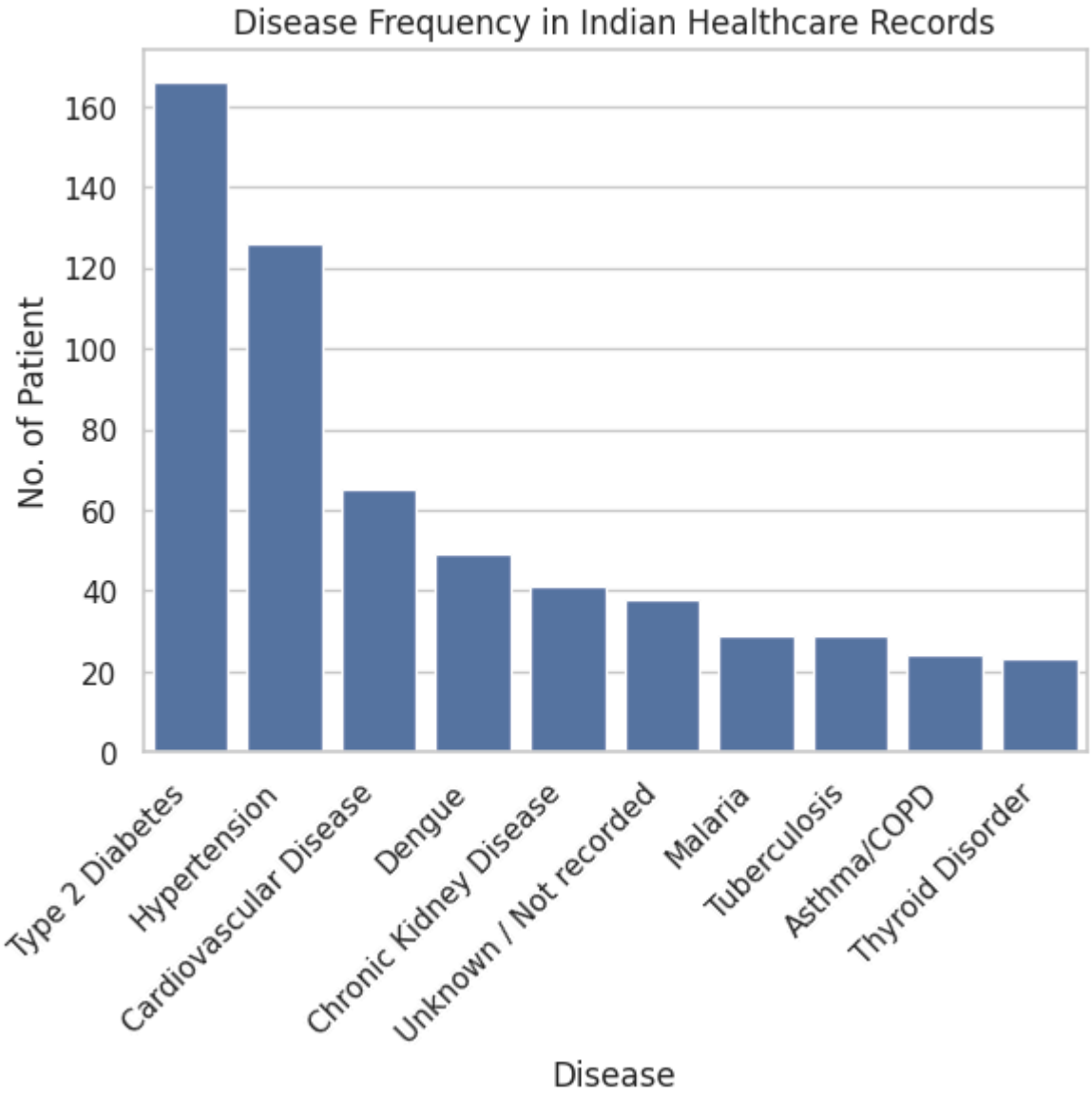
In [ ]:
```
import seaborn as sns
```

1. Which diseases are most frequently observed in our 590 healthcare records?

In [ ]:
```
most_diseases = pd.DataFrame(df['Primary_Diagnosis'].value_counts())
most_diseases.index.name = 'Disease'
most_diseases.columns = ['No. of Patient']
print(most_diseases)
```

```
                       No. of Patient
Disease
Type 2 Diabetes                   166
Hypertension                      126
Cardiovascular Disease             65
Dengue                             49
Chronic Kidney Disease             41
Unknown / Not recorded             38
Malaria                            29
Tuberculosis                       29
Asthma/COPD                        24
Thyroid Disorder                   23
```

In [ ]:
```
sns.set_theme(style = 'whitegrid')
sns.barplot(data=most_diseases, x='Disease', y='No. of Patient')
plt.xticks(rotation=45, ha='right')
plt.title('Disease Frequency in Indian Healthcare Records')
plt.show()
```

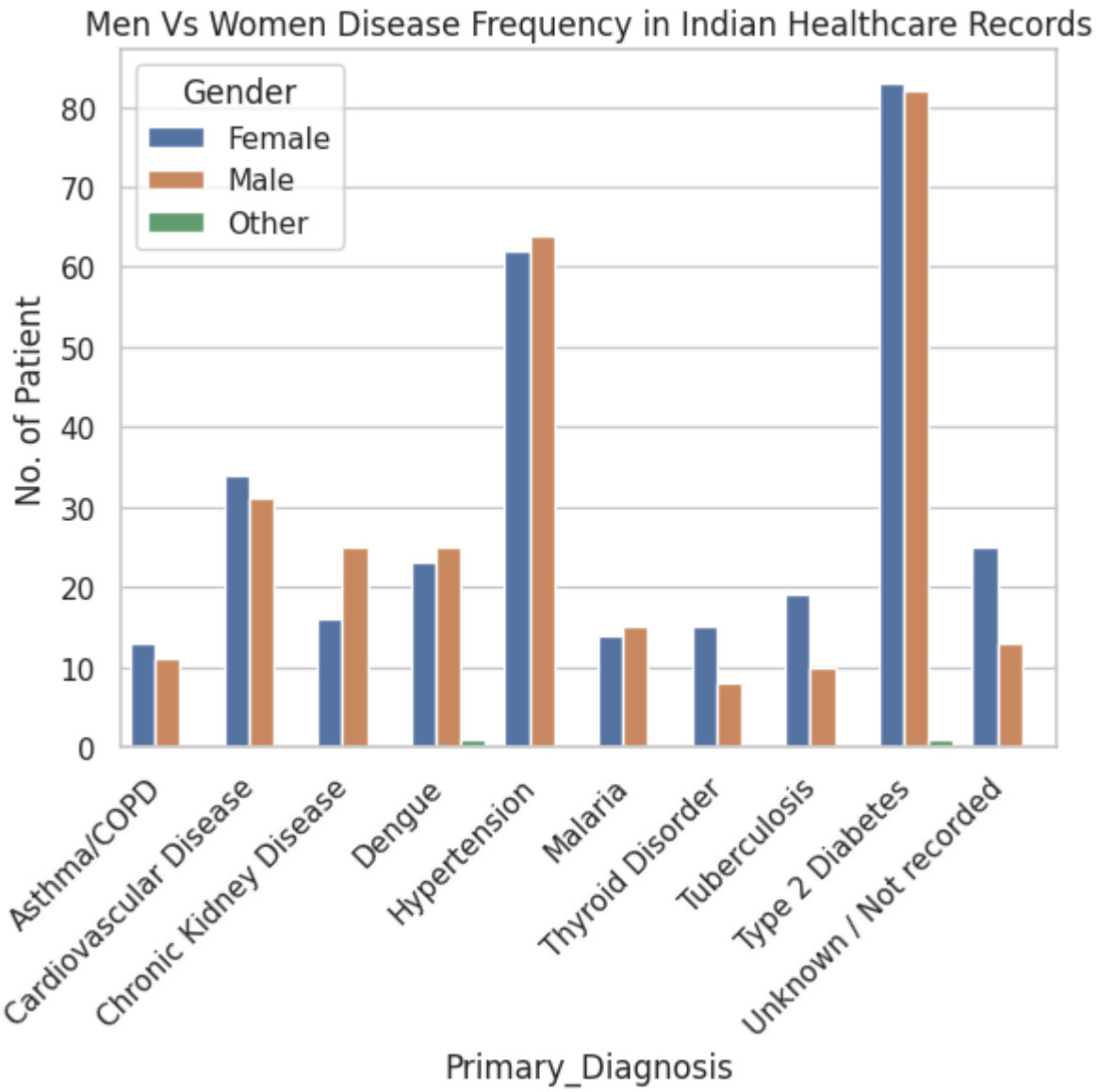


1. Different diseases ka distribution Male aur Female patients mein kaisa hai?

In [ ]:
```
disease_distribution = pd.DataFrame(
    df.groupby(['Primary_Diagnosis', 'Gender'])['Patient_ID']
      .count()
      .reset_index(name='No. of Patient')
)
print(disease_distribution)
```

```
         Primary_Diagnosis  Gender  No. of Patient
0              Asthma/COPD  Female              13
1              Asthma/COPD    Male              11
2   Cardiovascular Disease  Female              34
3   Cardiovascular Disease    Male              31
4   Chronic Kidney Disease  Female              16
5   Chronic Kidney Disease    Male              25
6                   Dengue  Female              23
7                   Dengue    Male              25
8                   Dengue   Other               1
9             Hypertension  Female              62
10            Hypertension    Male              64
11                 Malaria  Female              14
12                 Malaria    Male              15
13        Thyroid Disorder  Female              15
14        Thyroid Disorder    Male               8
15            Tuberculosis  Female              19
16            Tuberculosis    Male              10
17         Type 2 Diabetes  Female              83
18         Type 2 Diabetes    Male              82
19         Type 2 Diabetes   Other               1
20  Unknown / Not recorded  Female              25
21  Unknown / Not recorded    Male              13
```

In [ ]:
```
sns.set_theme(style = 'whitegrid')
sns.barplot(data=disease_distribution,x = 'Primary_Diagnosis',y='No. of
Patient',hue = 'Gender')
plt.title('Men Vs Women Disease Frequency in Indian Healthcare Records')
plt.xticks(rotation =45 , ha = 'right')
plt.show()
```



1. Regional Disease Distribution of Type 2 Diabetes Patients in India

In [ ]:
```
type_2_diabetes = df[df['Primary_Diagnosis']=='Type 2 Diabetes'].groupby('Region')['Patient_ID'].count().reset_index(name='No of Patient')
print(type_2_diabetes)
```

```
           Region  No of Patient
0  Andhra Pradesh             16
1           Delhi             19
2       Karnataka             24
3          Kerala             20
4     Maharashtra              7
5      Puducherry             13
6      Tamil Nadu             25
7       Telangana             14
8   Uttar Pradesh             17
9     West Bengal             11
```

In [ ]:
```
import folium
```

In [ ]:
```
m = folium.Map(location=[22.5, 80],zoom_start=5)
m
```

Out[309]: Make this Notebook Trusted to load map: File → Trust Notebook

In [ ]:
```
import geopandas as gpd
```

In [ ]:
```
india_map = gpd.read_file(
    '/content/Admin2.shp',
    on_invalid='fix'
)

india_map.head()
```

Out[311]:

	ST_NM	geometry
0	Arunachal Pradesh	POLYGON ((96.08866 29.45997, 96.09428 29.45477...
1	Assam	POLYGON ((95.97166 27.96254, 95.97174 27.96227...
2	Chandigarh	POLYGON ((76.77175 30.79498, 76.77231 30.7942,...
3	Karnataka	POLYGON ((77.32647 18.45884, 77.32648 18.45803...
4	Manipur	POLYGON ((94.57315 25.69156, 94.57522 25.69094...

In [ ]:
```
print(india_map.columns)
```

```
Index(['ST_NM', 'geometry'], dtype='object')
```

In [ ]:
```
print(india_map['ST_NM'].unique())
```

```
['Arunachal Pradesh' 'Assam' 'Chandigarh' 'Karnataka' 'Manipur'
 'Meghalaya' 'Mizoram' 'Nagaland' 'Punjab' 'Rajasthan' 'Sikkim' 'Tripura'
 'Uttarakhand' 'Telangana' 'Bihar' 'Kerala' 'Madhya Pradesh'
 'Andaman & Nicobar' 'Gujarat' 'Lakshadweep' 'Odisha'
 'Dadra and Nagar Haveli and Daman and Diu' 'Jammu & Kashmir'
 'Chhattisgarh' 'Delhi' 'Goa' 'Haryana' 'Himachal Pradesh' 'Jharkhand'
 'Tamil Nadu' 'Uttar Pradesh' 'West Bengal' 'Andhra Pradesh' 'Puducherry'
 'Maharashtra' 'Ladakh']
```

In [ ]:
```
diabetes_map = india_map[
    india_map['ST_NM'].isin(type_2_diabetes['Region'])
].merge(
    type_2_diabetes,
    left_on='ST_NM',
    right_on='Region',
    how='left'
)

diabetes_map[['ST_NM', 'No of Patient']]
```

Out[314]:

	ST_NM	No of Patient
0	Karnataka	24
1	Telangana	14
2	Kerala	20
3	Delhi	19
4	Tamil Nadu	25
5	Uttar Pradesh	17
6	West Bengal	11
7	Andhra Pradesh	16
8	Puducherry	13
9	Maharashtra	7

In [ ]:
```
m_diabetes_clean = folium.Map(
    location=[22.5, 80],
    zoom_start=5,
    tiles=None
)
```

In [ ]:
```
folium.Choropleth(
    geo_data=diabetes_map.__geo_interface__,
    data=diabetes_map,
    columns=['ST_NM', 'No of Patient'],
    key_on='feature.properties.ST_NM',
    fill_color='YlOrRd',
    legend_name='Recorded Type 2 Diabetes Patients'
).add_to(m_diabetes_clean)

m_diabetes_clean
```

Out[316]: Make this Notebook Trusted to load map: File → Trust Notebook

In [ ]:
```
m_diabetes_clean.save('/content/type2_diabetes_choropleth.html')
```

The choropleth map shows regional variation in the recorded Type 2 Diabetes patients within the dataset. Darker shades indicate relatively higher concentrations of recorded patients, while lighter shades indicate relatively lower concentrations.

1. Regional Disease Distribution of Hypertension Patients in India

In [ ]:
```
Hypertension = df[df['Primary_Diagnosis']=='Hypertension'].groupby('Region')
['Patient_ID'].count().reset_index(name='No of Patient')
print(Hypertension)
```

```
           Region  No of Patient
0  Andhra Pradesh             13
1           Delhi              8
2       Karnataka             16
3          Kerala             19
4     Maharashtra              8
5      Puducherry             12
6      Tamil Nadu             26
7       Telangana             11
8   Uttar Pradesh              8
9     West Bengal              5
```

In [ ]:
```
f = folium.Map(location=[22.5, 80],zoom_start=5)
f
```

Out[319]: Make this Notebook Trusted to load map: File → Trust Notebook

In [ ]:
```
Hypertension_map = india_map[
    india_map['ST_NM'].isin(Hypertension['Region'])
].merge(
    Hypertension,
    left_on='ST_NM',
    right_on='Region',
    how='left'
)

Hypertension_map[['ST_NM', 'No of Patient']]
```

Out[320]:

	ST_NM	No of Patient
0	Karnataka	16
1	Telangana	11
2	Kerala	19
3	Delhi	8
4	Tamil Nadu	26
5	Uttar Pradesh	8
6	West Bengal	5
7	Andhra Pradesh	13
8	Puducherry	12
9	Maharashtra	8

In [ ]:
```
f_hypertension_clean = folium.Map(
    location=[22.5, 80],
    zoom_start=5,
    tiles=None
)
```

In [ ]:
```
folium.Choropleth(
    geo_data=Hypertension_map.__geo_interface__,
    data=Hypertension_map,
    columns=['ST_NM', 'No of Patient'],
    key_on='feature.properties.ST_NM',
    fill_color='YlOrRd',
    legend_name='Recorded Hypertension Patients'
).add_to(f_hypertension_clean)

f_hypertension_clean
```

Out[322]: Make this Notebook Trusted to load map: File → Trust Notebook

In [ ]:
```
f_hypertension_clean.save('/content/hypertension_choropleth.html')
```

1. Disease Frequency Across Age Groups

In [ ]:
```
df['Age_group'] = pd.cut(df['Age'],bins=[0,18,35,50,65,float('inf')],
                         labels = ['0-18','19-35','36-50','51-65','66+'],
                         include_lowest = True)
```

In [ ]:
```
print(df[['Age', 'Age_group']].head(10))
```

```
   Age Age_group
0   24     19-35
1   63     51-65
2   38     36-50
3   58     51-65
4   54     51-65
5   36     36-50
6   54     51-65
7   24     19-35
8   51     51-65
9   36     36-50
```

In [ ]:
```
print(df['Age_group'].value_counts().sort_index())
```

```
Age_group
0-18      50
19-35    134
36-50    184
51-65    153
66+       69
Name: count, dtype: int64
```

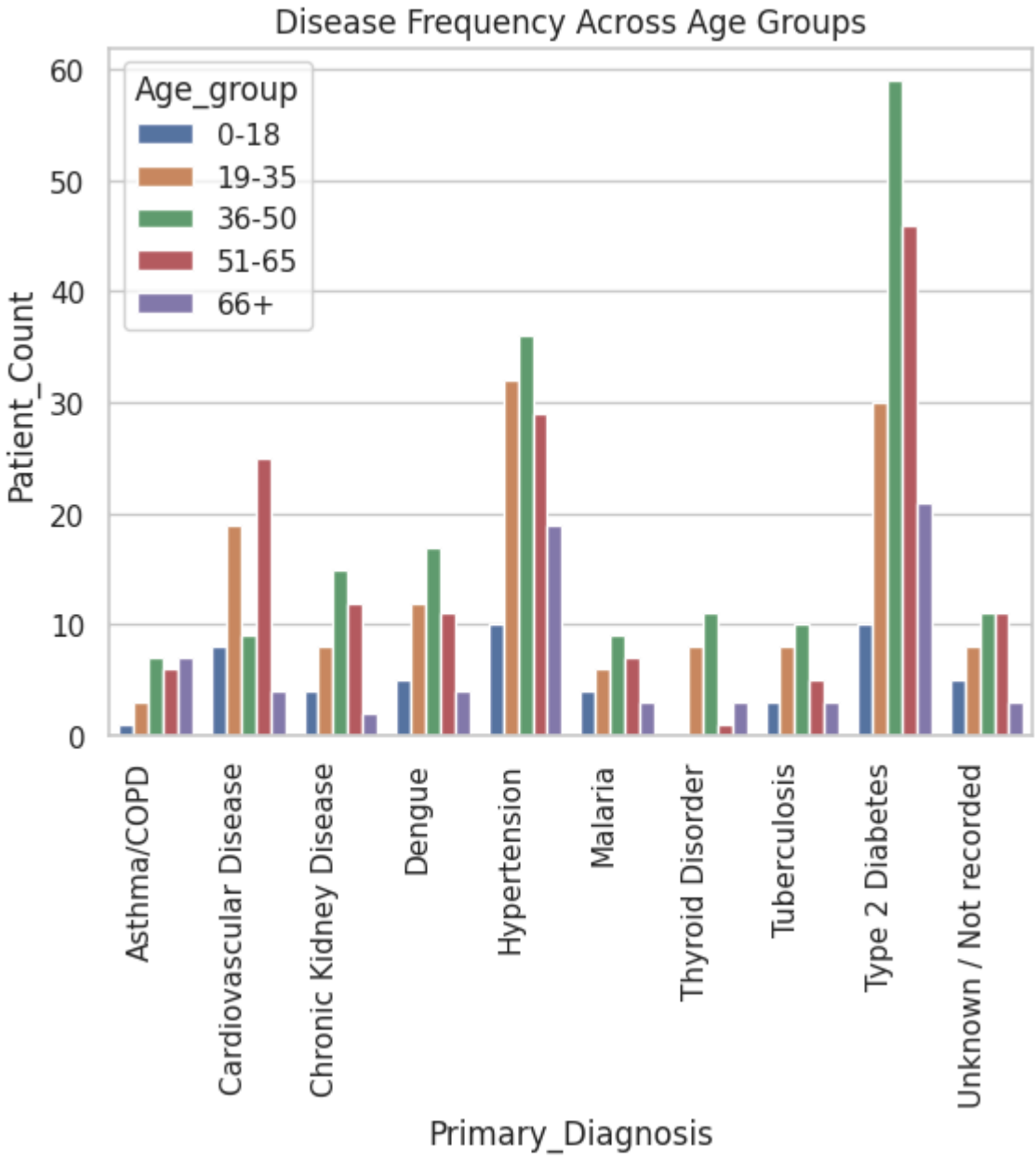
In [ ]:
```
age_disease = (
    df.groupby(['Age_group', 'Primary_Diagnosis'])['Patient_ID']
    .count()
    .reset_index(name='Patient_Count')
)
print(age_disease)
```

	Age_group	Primary_Diagnosis	Patient_Count
0	0-18	Asthma/COPD	1
1	0-18	Cardiovascular Disease	8
2	0-18	Chronic Kidney Disease	4
3	0-18	Dengue	5
4	0-18	Hypertension	10
5	0-18	Malaria	4
6	0-18	Thyroid Disorder	0
7	0-18	Tuberculosis	3
8	0-18	Type 2 Diabetes	10
9	0-18	Unknown / Not recorded	5
10	19-35	Asthma/COPD	3
11	19-35	Cardiovascular Disease	19
12	19-35	Chronic Kidney Disease	8
13	19-35	Dengue	12
14	19-35	Hypertension	32
15	19-35	Malaria	6
16	19-35	Thyroid Disorder	8
17	19-35	Tuberculosis	8
18	19-35	Type 2 Diabetes	30
19	19-35	Unknown / Not recorded	8
20	36-50	Asthma/COPD	7
21	36-50	Cardiovascular Disease	9
22	36-50	Chronic Kidney Disease	15
23	36-50	Dengue	17
24	36-50	Hypertension	36
25	36-50	Malaria	9
26	36-50	Thyroid Disorder	11
27	36-50	Tuberculosis	10
28	36-50	Type 2 Diabetes	59
29	36-50	Unknown / Not recorded	11
30	51-65	Asthma/COPD	6
31	51-65	Cardiovascular Disease	25
32	51-65	Chronic Kidney Disease	12
33	51-65	Dengue	11
34	51-65	Hypertension	29
35	51-65	Malaria	7
36	51-65	Thyroid Disorder	1
37	51-65	Tuberculosis	5
38	51-65	Type 2 Diabetes	46
39	51-65	Unknown / Not recorded	11
40	66+	Asthma/COPD	7
41	66+	Cardiovascular Disease	4
42	66+	Chronic Kidney Disease	2
43	66+	Dengue	4
44	66+	Hypertension	19

```
45       66+                  Malaria              3
46       66+          Thyroid Disorder             3
47       66+              Tuberculosis             3
48       66+           Type 2 Diabetes            21
49       66+  Unknown / Not recorded              3
```

```
/tmp/ipykernel_3262/3113736921.py:2: FutureWarning: The default of
observed=False is deprecated and will be changed to True in a future version
of pandas. Pass observed=False to retain current behavior or observed=True
to adopt the future default and silence this warning.
  df.groupby(['Age_group', 'Primary_Diagnosis'])['Patient_ID']
```

In [ ]:
```
sns.set_theme(style = 'whitegrid')
sns.barplot(data=age_disease,x = 'Primary_Diagnosis',y='Patient_Count',hue =
'Age_group')
plt.title('Disease Frequency Across Age Groups')
plt.xticks(rotation =90 , ha = 'right')
plt.show()
```



1. Disease vs Treatment Outcome

In [ ]:
```
outcomes = (df.groupby(['Primary_Diagnosis','Treatment_Outcome'])
['Patient_ID'].count().reset_index(name='Patient_Count'))
print(outcomes)
```

	Primary_Diagnosis	Treatment_Outcome	Patient_Count
0	Asthma/COPD	Improved	2
1	Asthma/COPD	Recovered	4
2	Asthma/COPD	Referred	2
3	Asthma/COPD	Stable	4
4	Asthma/COPD	Unknown	1
5	Asthma/COPD	Worsened	11
6	Cardiovascular Disease	Improved	16
7	Cardiovascular Disease	Recovered	14
8	Cardiovascular Disease	Referred	10
9	Cardiovascular Disease	Stable	12
10	Cardiovascular Disease	Unknown	3
11	Cardiovascular Disease	Worsened	10
12	Chronic Kidney Disease	Improved	9
13	Chronic Kidney Disease	Recovered	9
14	Chronic Kidney Disease	Referred	6
15	Chronic Kidney Disease	Stable	4
16	Chronic Kidney Disease	Unknown	2
17	Chronic Kidney Disease	Worsened	11
18	Dengue	Improved	8
19	Dengue	Recovered	8
20	Dengue	Referred	16
21	Dengue	Stable	7
22	Dengue	Unknown	3
23	Dengue	Worsened	7
24	Hypertension	Improved	28
25	Hypertension	Recovered	22
26	Hypertension	Referred	33
27	Hypertension	Stable	24
28	Hypertension	Unknown	1
29	Hypertension	Worsened	18
30	Malaria	Improved	5
31	Malaria	Recovered	6
32	Malaria	Referred	4
33	Malaria	Stable	4
34	Malaria	Worsened	10
35	Thyroid Disorder	Improved	4
36	Thyroid Disorder	Recovered	5
37	Thyroid Disorder	Referred	4
38	Thyroid Disorder	Stable	5
39	Thyroid Disorder	Worsened	5
40	Tuberculosis	Improved	4
41	Tuberculosis	Recovered	11
42	Tuberculosis	Referred	5
43	Tuberculosis	Stable	7
44	Tuberculosis	Worsened	2
45	Type 2 Diabetes	Improved	27
46	Type 2 Diabetes	Recovered	30
47	Type 2 Diabetes	Referred	35

```
48          Type 2 Diabetes            Stable             33
49          Type 2 Diabetes           Unknown              5
50          Type 2 Diabetes          Worsened             36
51  Unknown / Not recorded          Improved              8
52  Unknown / Not recorded         Recovered              6
53  Unknown / Not recorded          Referred             11
54  Unknown / Not recorded            Stable              7
55  Unknown / Not recorded          Worsened              6
```

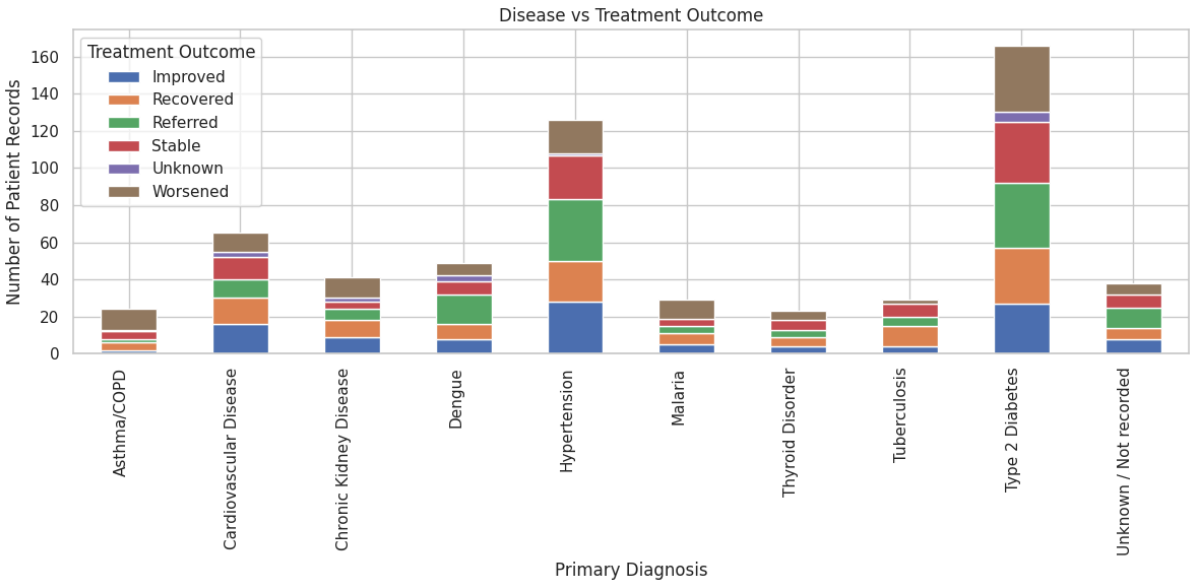
In [ ]:

```
sns.set_theme(style='whitegrid')

outcomes_pivot = outcomes.pivot(
    index='Primary_Diagnosis',
    columns='Treatment_Outcome',
    values='Patient_Count'
).fillna(0)

outcomes_pivot.plot(
    kind='bar',
    stacked=True,
    figsize=(12, 6)
)

plt.title('Disease vs Treatment Outcome')
plt.xlabel('Primary Diagnosis')
plt.ylabel('Number of Patient Records')
plt.xticks(rotation=90, ha='right')
plt.legend(title='Treatment Outcome')
plt.tight_layout()
plt.show()
```



1. Disease Frequency Across BMI Categories

In [ ]:
```
df['BMI_Group'] = pd.cut(df['BMI'],bins=[0,18,25,30,float('inf')],
                         labels =
['Underweight','Normal','Overweight','Obese'],
                         include_lowest = True)
```

In [ ]:
```
print(df[['BMI', 'BMI_Group']].head(10))
```

```
    BMI    BMI_Group
0  38.8        Obese
1  26.0   Overweight
2  20.4       Normal
3  22.4       Normal
4  17.6  Underweight
5  28.4   Overweight
6  31.0        Obese
7  25.2   Overweight
8  26.2   Overweight
9  24.5       Normal
```

In [ ]:
```
print(df['BMI_Group'].value_counts().sort_index())
```

```
BMI_Group
Underweight     69
Normal         257
Overweight     160
Obese          104
Name: count, dtype: int64
```

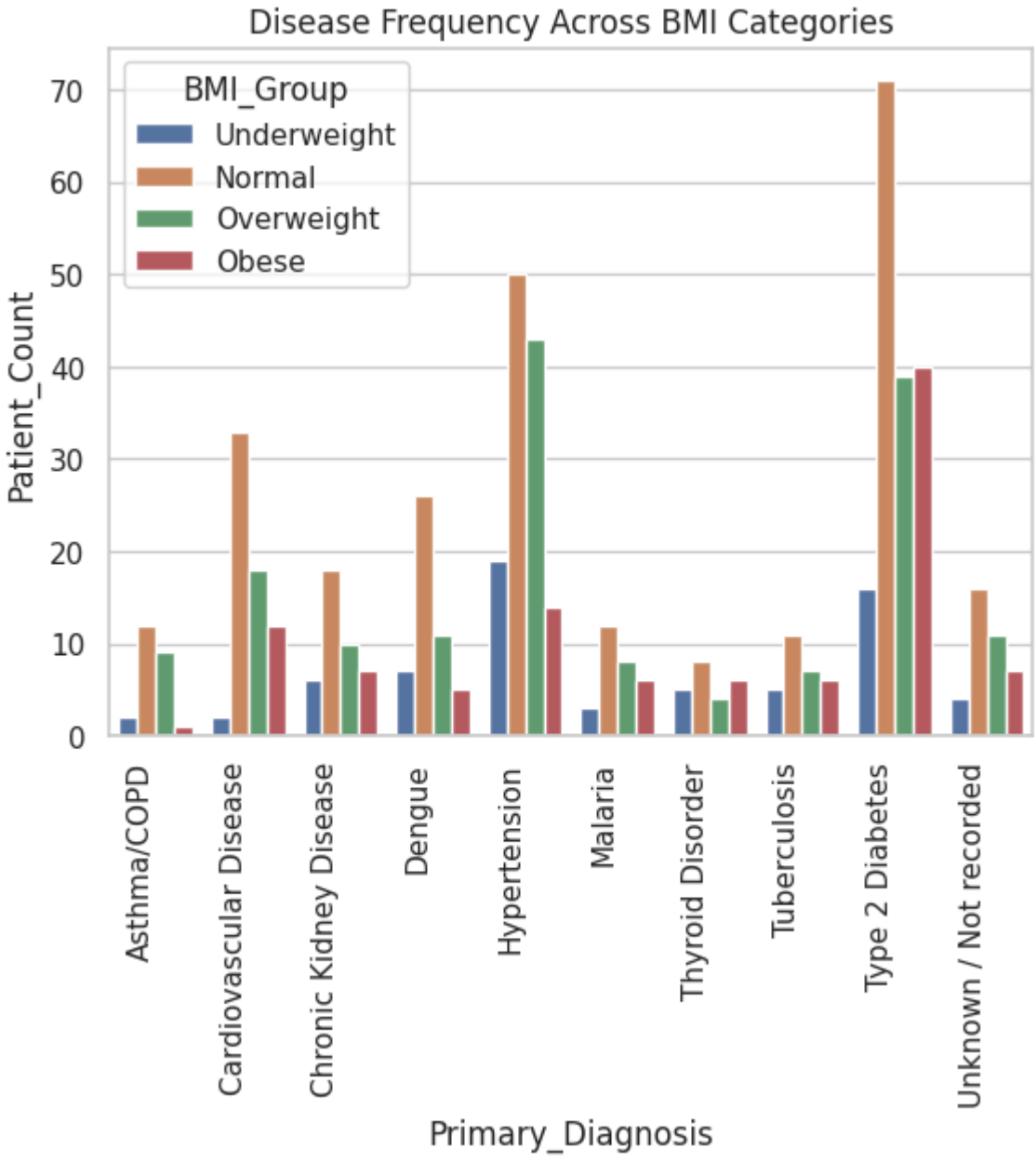
In [ ]:
```
BMI_disease = (
    df.groupby(['BMI_Group', 'Primary_Diagnosis'])['Patient_ID']
    .count()
    .reset_index(name='Patient_Count')
)
print(BMI_disease)
```

	BMI_Group	Primary_Diagnosis	Patient_Count
0	Underweight	Asthma/COPD	2
1	Underweight	Cardiovascular Disease	2
2	Underweight	Chronic Kidney Disease	6
3	Underweight	Dengue	7
4	Underweight	Hypertension	19
5	Underweight	Malaria	3
6	Underweight	Thyroid Disorder	5
7	Underweight	Tuberculosis	5
8	Underweight	Type 2 Diabetes	16
9	Underweight	Unknown / Not recorded	4
10	Normal	Asthma/COPD	12
11	Normal	Cardiovascular Disease	33
12	Normal	Chronic Kidney Disease	18
13	Normal	Dengue	26
14	Normal	Hypertension	50
15	Normal	Malaria	12
16	Normal	Thyroid Disorder	8
17	Normal	Tuberculosis	11
18	Normal	Type 2 Diabetes	71
19	Normal	Unknown / Not recorded	16
20	Overweight	Asthma/COPD	9
21	Overweight	Cardiovascular Disease	18
22	Overweight	Chronic Kidney Disease	10
23	Overweight	Dengue	11
24	Overweight	Hypertension	43
25	Overweight	Malaria	8
26	Overweight	Thyroid Disorder	4
27	Overweight	Tuberculosis	7
28	Overweight	Type 2 Diabetes	39
29	Overweight	Unknown / Not recorded	11
30	Obese	Asthma/COPD	1
31	Obese	Cardiovascular Disease	12
32	Obese	Chronic Kidney Disease	7
33	Obese	Dengue	5
34	Obese	Hypertension	14
35	Obese	Malaria	6
36	Obese	Thyroid Disorder	6
37	Obese	Tuberculosis	6
38	Obese	Type 2 Diabetes	40
39	Obese	Unknown / Not recorded	7

```
/tmp/ipykernel_3262/4261759462.py:2: FutureWarning: The default of
observed=False is deprecated and will be changed to True in a future version
of pandas. Pass observed=False to retain current behavior or observed=True
```

```
to adopt the future default and silence this warning.
  df.groupby(['BMI_Group', 'Primary_Diagnosis'])['Patient_ID']
```

In [ ]:
```
sns.set_theme(style = 'whitegrid')
sns.barplot(data=BMI_disease,x = 'Primary_Diagnosis',y='Patient_Count',hue =
'BMI_Group')
plt.title('Disease Frequency Across BMI Categories')
plt.xticks(rotation =90 , ha = 'right')
plt.show()
```



1. Clinical Indicators vs Disease

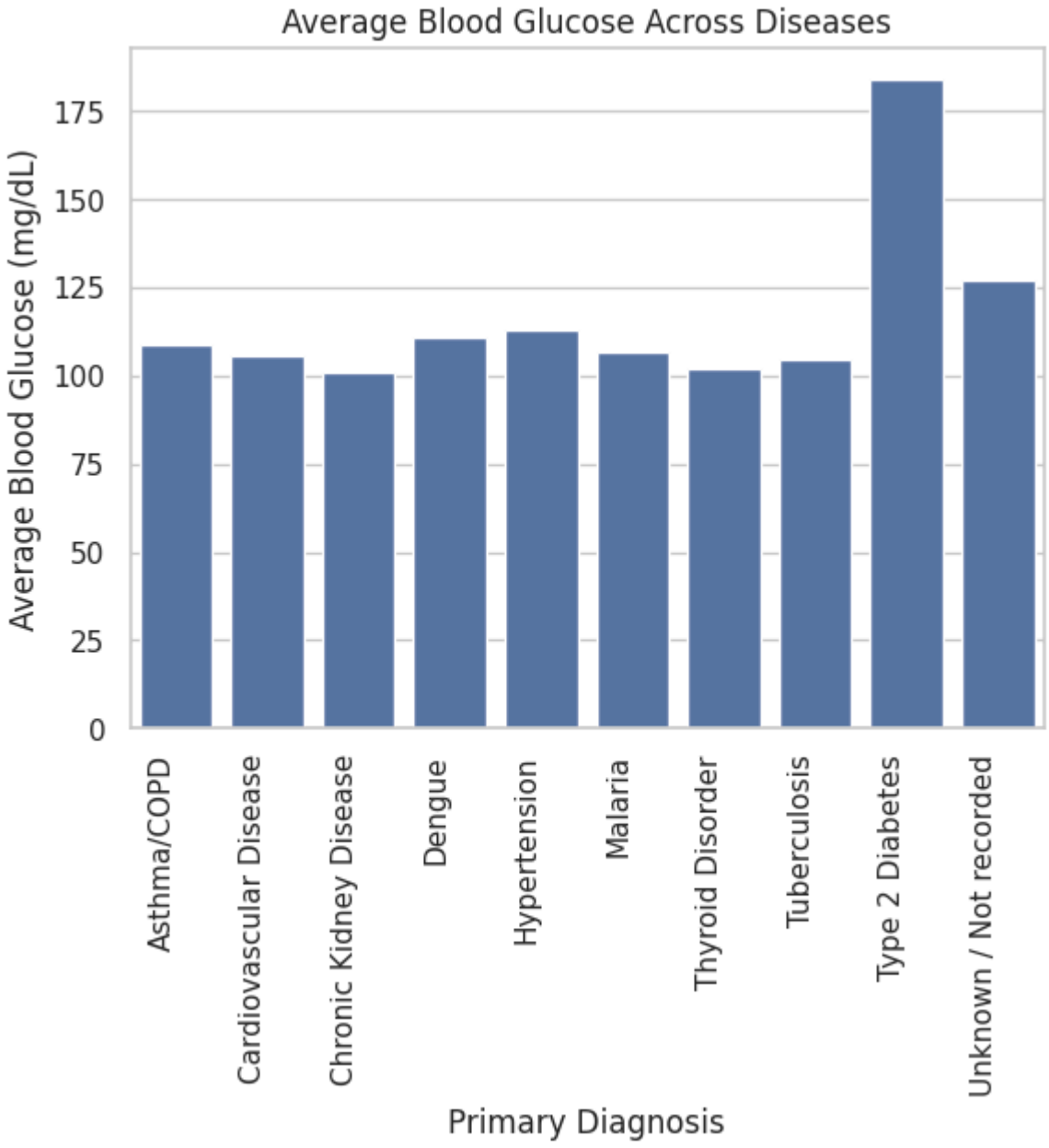
In [ ]:
```
clinical_summary = (df.groupby('Primary_Diagnosis').agg(
        Average_Glucose=('Blood_Glucose_mg_dL', 'mean'),
        Average_HbA1c=('HbA1c_%', 'mean')
    ).reset_index()
)
```

In [ ]:
```
print(clinical_summary.round(2))
```

	Primary_Diagnosis	Average_Glucose	Average_HbA1c
0	Asthma/COPD	108.72	5.55
1	Cardiovascular Disease	105.90	5.75
2	Chronic Kidney Disease	101.05	5.41
3	Dengue	110.81	5.73
4	Hypertension	112.98	5.59
5	Malaria	106.72	5.70
6	Thyroid Disorder	101.87	5.47
7	Tuberculosis	104.69	5.61
8	Type 2 Diabetes	183.75	8.15
9	Unknown / Not recorded	127.09	6.84

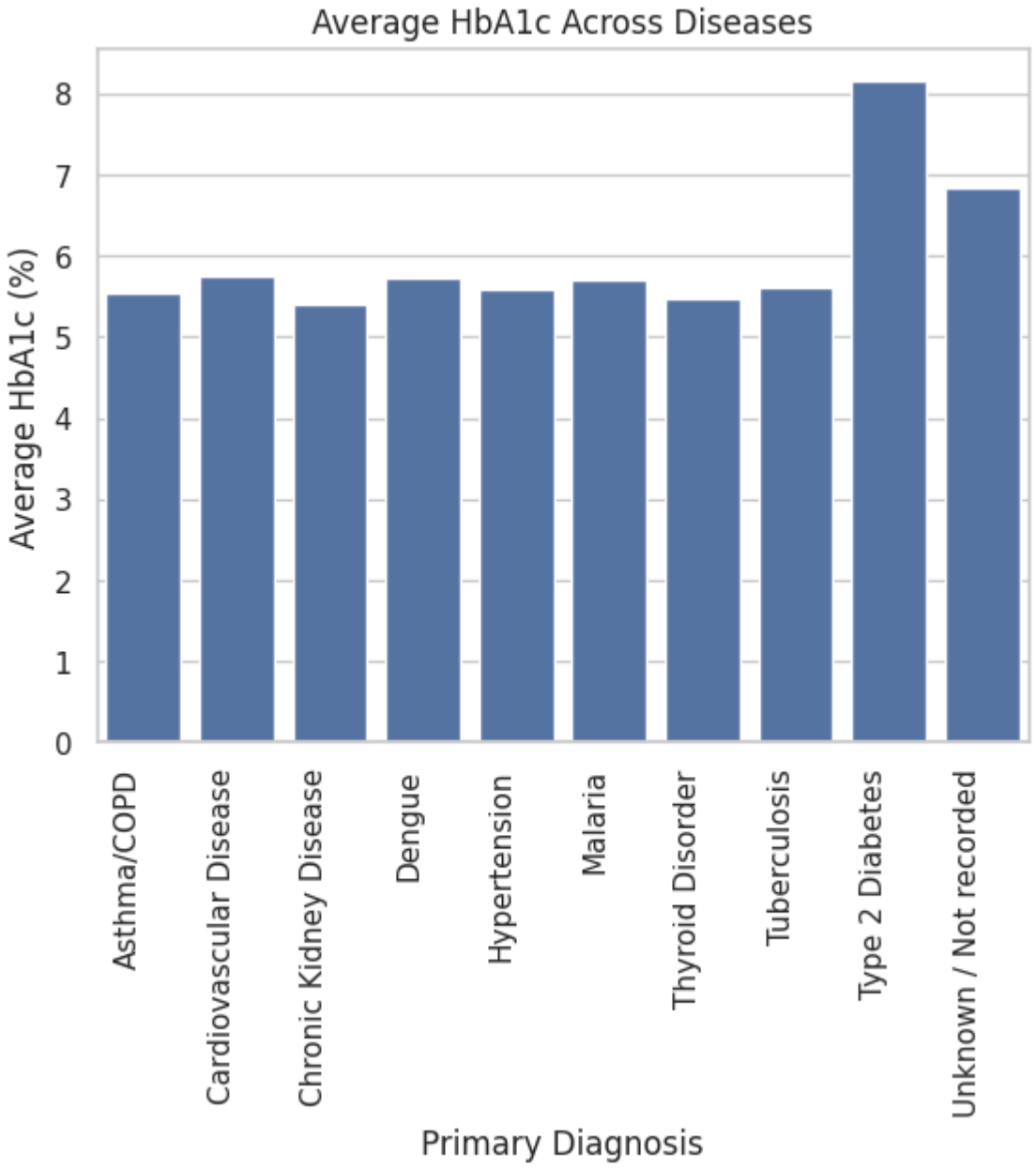
Graph 1: Average Blood Glucose

In [ ]:
```
sns.set_theme(style='whitegrid')
sns.barplot(data=clinical_summary,x='Primary_Diagnosis',y='Average_Glucose')
plt.title('Average Blood Glucose Across Diseases')
plt.xlabel('Primary Diagnosis')
plt.ylabel('Average Blood Glucose (mg/dL)')
plt.xticks(rotation=90, ha='right')
plt.show()
```



Graph 2: Average HbA1c

In [ ]:
```
sns.set_theme(style='whitegrid')
sns.barplot(data=clinical_summary,x='Primary_Diagnosis',y='Average_HbA1c')
plt.title('Average HbA1c Across Diseases')
plt.xlabel('Primary Diagnosis')
plt.ylabel('Average HbA1c (%)')
plt.xticks(rotation=90, ha='right')
plt.show()
```



1. Correlation Analysis

In [ ]:
```
dff =
pd.DataFrame(df[['Age','Blood_Glucose_mg_dL','HbA1c_%','Total_Cholesterol_mg
_dL','BMI']])
corr = dff.corr(numeric_only = True)
print(corr)
```

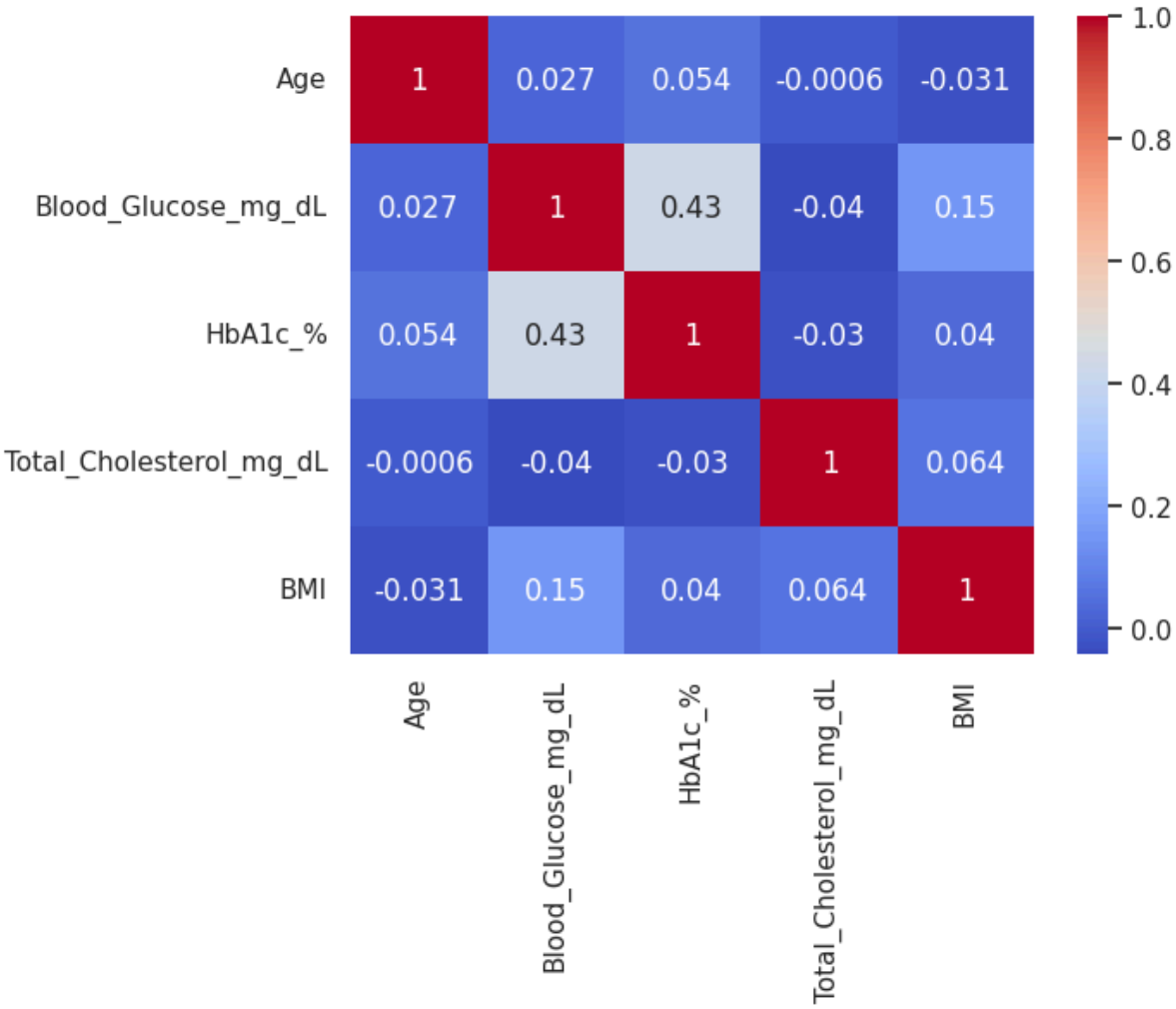
```
                              Age  Blood_Glucose_mg_dL   HbA1c_%  \
Age                      1.000000             0.027169  0.053602
Blood_Glucose_mg_dL      0.027169             1.000000  0.432124
HbA1c_%                  0.053602             0.432124  1.000000
Total_Cholesterol_mg_dL -0.000604            -0.040326 -0.029586
BMI                     -0.031238             0.150347  0.039861

                         Total_Cholesterol_mg_dL       BMI
Age                                    -0.000604 -0.031238
Blood_Glucose_mg_dL                    -0.040326  0.150347
HbA1c_%                                -0.029586  0.039861
Total_Cholesterol_mg_dL                 1.000000  0.063698
BMI                                     0.063698  1.000000
```

To visualize an entire correlation matrix we are going to use Heatmap using seaborn

In [ ]:
```
sns.heatmap(corr , annot = True , cmap = 'coolwarm')
```

Out[341]: <Axes: >



1. Monthly Disease Frequency Analysis

In [ ]:
```
df['Visit_Date'] = pd.to_datetime(df['Visit_Date'],errors='coerce')
print(df['Visit_Date'].dtype)
```

```
datetime64[ns]
```

In [ ]:
```
print("Earliest Visit Date:", df['Visit_Date'].min())
print("Latest Visit Date:", df['Visit_Date'].max())
print("Missing Visit Dates:", df['Visit_Date'].isnull().sum())
```

```
Earliest Visit Date: 2024-05-28 00:00:00
Latest Visit Date: 2026-05-25 00:00:00
Missing Visit Dates: 8
```

In [ ]:
```
df['Visit_Month'] = df['Visit_Date'].dt.to_period('M').astype(str)
print(df[['Visit_Date', 'Visit_Month']].head(10))
```

```
  Visit_Date Visit_Month
0 2024-11-12     2024-11
1 2025-02-17     2025-02
2 2025-06-12     2025-06
3 2025-06-30     2025-06
4 2024-09-16     2024-09
5 2024-11-22     2024-11
6 2024-07-06     2024-07
7 2026-01-11     2026-01
8 2025-02-17     2025-02
9 2025-07-10     2025-07
```

In [ ]:
```
monthly_records = (
    df.dropna(subset=['Visit_Date'])
    .groupby('Visit_Month')['Patient_ID']
    .count()
    .reset_index(name='Patient_Count')
)

print(monthly_records)
```

```
   Visit_Month  Patient_Count
0      2024-05              3
1      2024-06             24
2      2024-07             34
3      2024-08             34
4      2024-09             31
5      2024-10             33
6      2024-11             12
7      2024-12             29
8      2025-01             19
9      2025-02             20
10     2025-03             17
11     2025-04             25
12     2025-05             26
13     2025-06             27
14     2025-07             18
15     2025-08             24
16     2025-09             20
17     2025-10             25
18     2025-11             30
19     2025-12             23
20     2026-01             18
21     2026-02             16
22     2026-03             21
23     2026-04             27
24     2026-05             26
```

In [ ]:
```
df.to_csv("new_cleaned_healthcare_data.csv", index=False)
```